

## Phase A — technical validity

synthetic smoke run + engine invariants + per-family leakage tests  
(prefix invariance, correct lag, next-bar execution, warm-up)  
says nothing about profitability

*gate: invariants + leakage tests pass*

## Phase B — development pilot

2 assets × 1 seed × 15 folds · 60 evaluations per fold and engine  
pipeline correctness only — pilot Sharpe is never quotable evidence  
(R3 pilots: all 5 families advanced on process grounds only)

*gate: futility check only — no evidence*

## Phase C — full study

2 assets × 10 seeds × 15 folds · budget 100/fold/engine  
full statistical battery (Ch. 6) · promotion gate C1-C6  
R3 outcome: 5 in → 0 promoted (CLOSED\_NEGATIVE)

*gate: C1-C6, each on ≥6/10 seeds (Table 3.2)*

candidate status → single confirmatory holdout shot (§3.4)